

ECO 362: Financial Investments Taxes

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2025 income tax rates for single filers

Taxable income (\$)		Marginal rate (%)
Min	Max	
0	11,000	10
11,001	44,725	12
44,726	95,375	22
95,376	182,100	24
182,101	231,250	32
231,251	578,125	35
578,126		37

2025 capital gains tax rates for single filers

Taxable income (\$)		Rate (%)
Min	Max	
0	48,350	0
48,351	533,400	15
533,401		20

- Long-term capital gains tax rates for assets held more than a year.
- Short-term capital gains (for asset held less than a year) are taxed at the income tax rate.

Tax advantaged accounts

- 401(k) or 403(b) account through your employer.
 - Traditional or Roth.
 - Contribution limit in 2025: \$23,500 (\$31,000 if 50 or older).
- Individual Retirement Account (IRA).
 - Traditional or Roth.
 - Contribution limit in 2025: \$7,000 (\$8,000 if 50 or older).
- 529 account.
 - Annual gift tax exemption of \$19,000 per person.
 - Economically similar to a Roth account if withdrawals are used for educational expenses.

Traditional vs. Roth accounts

- Traditional account.
 - Contributions (up to the limit) are tax deductible.
 - Tax-free accumulation until withdrawal.
 - All withdrawals are taxed at the income tax rate.
- Roth account.
 - Contributions are from after-tax income.
 - Tax-free accumulation.
 - No taxes on withdrawals.

Notation

- Let τ_i be the income tax rate.
- Let τ_c be the capital gains tax rate.
- Let the return before tax be

$$\begin{aligned} R_t &= \frac{P_t + D_t}{P_{t-1}} - 1 \\ &= \underbrace{\frac{P_t}{P_{t-1}} - 1}_{\text{capital gain}} + \underbrace{\frac{D_t}{P_{t-1}}}_{\text{dividend yield}} \end{aligned}$$

where

- P_t is price.
- D_t is dividend or coupon.

After-tax return in a taxable account

- After-tax return if capital gain is unrealized.

$$R_t = \frac{P_t}{P_{t-1}} - 1 + \frac{(1 - \tau_i)D_t}{P_{t-1}}$$

- After-tax return if capital gain is realized.

$$R_t = (1 - \tau_c) \left(\frac{P_t}{P_{t-1}} - 1 \right) + \frac{(1 - \tau_i)D_t}{P_{t-1}}$$

- Leave capital gains unrealized, so that your investment compounds tax-free.

Example 1: Wealth under 3 strategies

- Start with \$10,000 of pre-tax income.
- Investment horizon of $T = 30$ years.
- Annual riskless interest of $R = 3\%$.
- Marginal income tax rate of $\tau_i = 24\%$, assumed to remain constant in the future.

1. Taxable account.

$$\begin{aligned}(1 - \tau_i)\$10,000 \times (1 + (1 - \tau_i)R)^T \\&= (1 - 0.24)\$10,000 \times (1 + (1 - 0.24)0.03)^{30} \\&= \$14,946\end{aligned}$$

2. Traditional 401(k).

$$\begin{aligned}\$10,000(1 + R)^T \times (1 - \tau_i) \\&= \$10,000(1.03)^{30} \times (1 - 0.24) \\&= \$18,447\end{aligned}$$

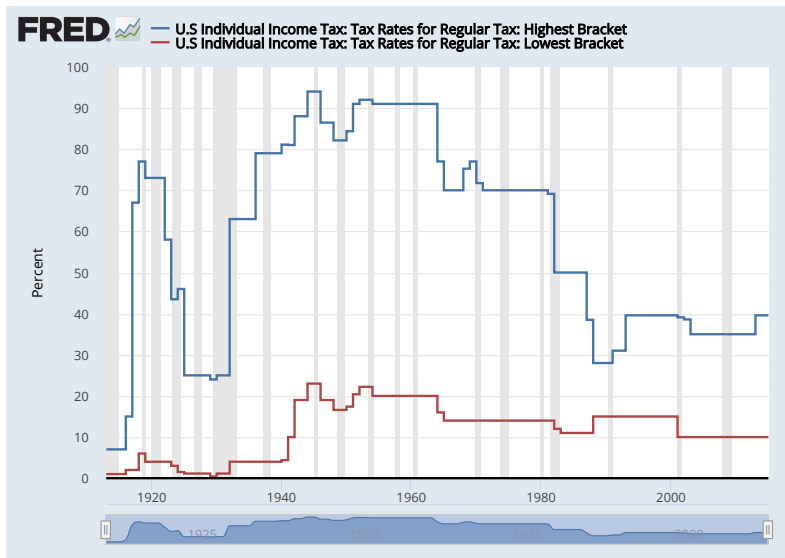
3. Roth 401(k).

$$\begin{aligned}(1 - \tau_i)\$10,000 \times (1 + R)^T \\&= (1 - 0.24)\$10,000 \times (1.03)^{30} \\&= \$18,447\end{aligned}$$

Traditional vs. Roth account

- Income tax paid at withdrawal in a traditional account.
- Income tax paid upfront in a Roth account.
- If the income tax rate is constant over time, the two accounts have the same return.
- If the income tax rate is expected to increase, the Roth account is more tax advantaged.
- In reality, not clear which account is more tax advantaged.
 - If income is lower in retirement, you will be in a lower tax bracket.
 - Congress may increase the tax rate in the future.

US income tax rates



Source: U.S. Department of the Treasury. Internal Revenue Service

Cumulative return with changing tax rates

- Taxable account.

$$(1 - \tau_{i,0}) \prod_{t=1}^T (1 + (1 - \tau_{i,t})R_t)$$

- Traditional 401(k).

$$(1 - \tau_{i,T}) \prod_{t=1}^T (1 + R_t)$$

- Roth 401(k).

$$(1 - \tau_{i,0}) \prod_{t=1}^T (1 + R_t)$$

Maximum tax shelter in a traditional account

- \$30,921 of before-tax income that you want to save.
- Marginal income tax rate of 24%.
- Save \$23,500 in a traditional 401(k) account.
- Save the remaining \$7,421 in a taxable account.
- Cannot shelter all of your savings because of the annual limit on contributions.

Maximum tax shelter in a Roth account

- Contributions to a Roth account are after-tax.
- First pay income taxes on \$30,921.

$$(1 - 0.24)\$30,921 = \$23,500$$

- Save \$23,500 in a Roth 401(k) account, exactly equal to the contribution limit.
- All of your savings are tax sheltered.
 - An after-tax limit of \$23,500 in a Roth account is more generous than a pre-tax limit of \$23,500 in a traditional account.
- Favor a Roth account if you have the choice and intend to make the maximum contribution.

Asset allocation vs. location

- Asset allocation: Which assets should you hold?
- Asset location: Which account should you hold the assets in?
- Difficult problem, but some simple rules-of-thumb.
 - Hold taxable bonds and high-dividend stocks in tax advantaged accounts.
 - Consider tax-exempt (municipal) bonds and low-dividend stocks in taxable accounts.
 - If you need to rebalance (and realize capital gains), do so in a tax advantaged account.

Avoiding capital gains tax

- Capital gains on one asset can be offset by capital losses on another asset.
- In each tax year, capital losses up to \$3,000 can be deducted from ordinary income.
- Capital losses can be carried forward indefinitely to future tax years.

Tax loss harvesting

- Instead of buying a total market ETF, buy a subset of stocks (e.g., 500 stocks) that give you sufficient diversification.
- When a stock goes down substantially, sell it to claim a capital loss.
- Replace it with a different stock, avoiding wash sales.
 - Cannot sell then buy the same or a “substantially identical” security within 30 days.
- Direct indexing: Algorithms implement optimal tax loss harvesting in exchange for a fee (0.40% at Fidelity and Schwab).
 - Can generate tax alpha if you have substantial capital gains to offset (e.g., sale of a concentrated position or private business).

Summary

- Do not realize capital gains unless you have to.
- Use tax advantaged accounts.
- Traditional or Roth?
 - Not obvious because of uncertainty in future tax rates.
 - However, a Roth account allows you to shelter more savings.
- Direct indexing if you have substantial capital gains to offset.